



Scala Programming Practical

Practical – 4

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1. Generate a random data set of 10 numbers and calculate its variance and standard variance

Code:

```
import scala.util.Random

object Practical extends App {
  val data = Array.fill(10) (Random.nextInt(100))
  println("Random Dataset:" + data.mkString(", "))

  val mean = data.sum / data.length
  val s = data.map(x => math.pow(x - mean, 2))
  val v = s.sum / data.length
  val d = math.sqrt(v)

  println(f"Mean: $mean")
  println(f"Variance: $v%.2f")
  println(f"Standard Deviation: $d%.2f")
  println("Himanshu S037")
}
```

Output:

Random Dataset: 82, 45, 66, 29, 61, 95, 99, 8, 14, 70

Mean: 56

Variance: 928.50

Standard Deviation: 30.47

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